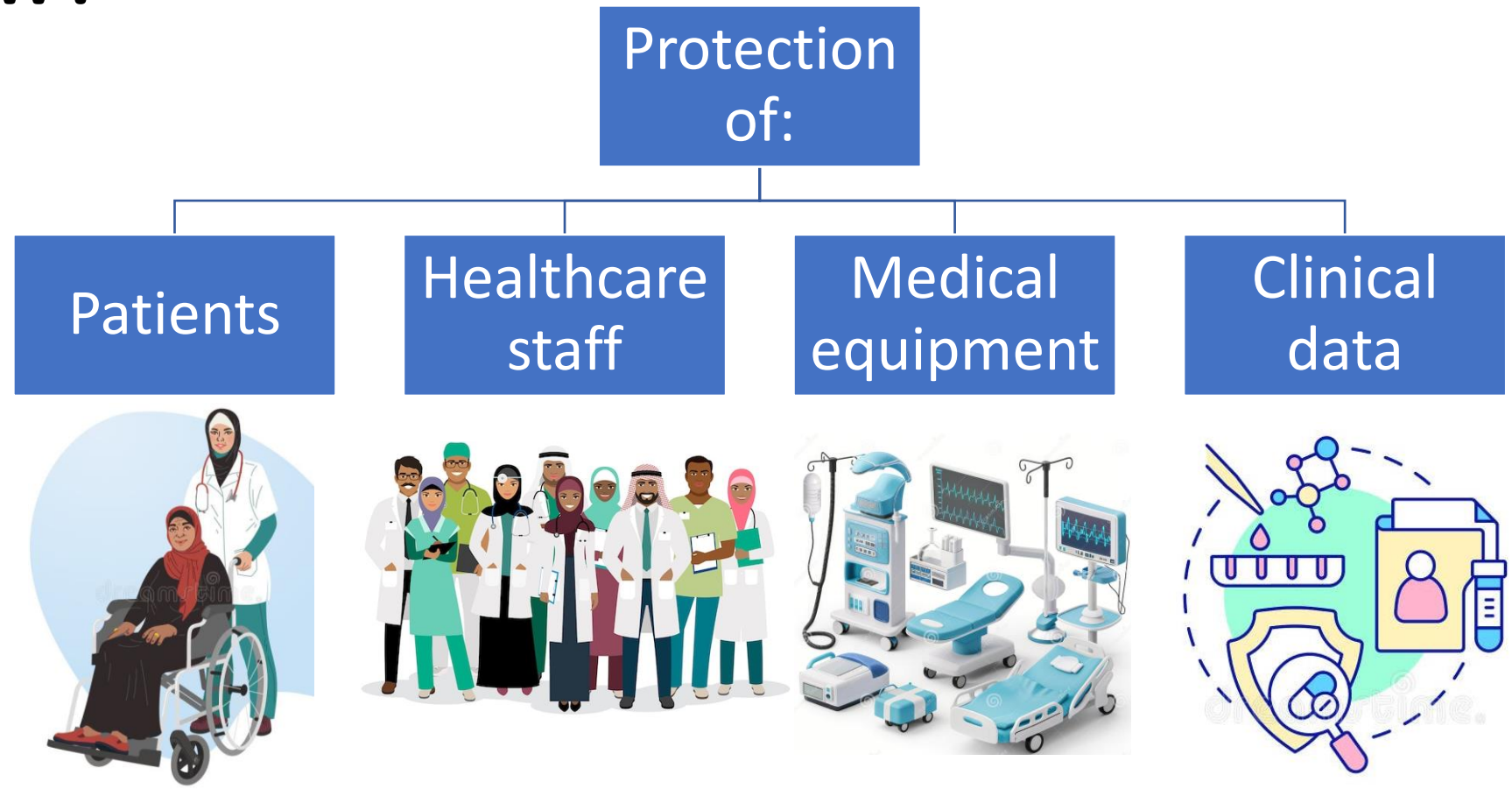
Lecture 5
Hospital Safety Systems

How Engineering Systems and Clinical Engineers Protect Patients

Engineering and Clinical Engineering Perspective

- Safety in hospitals is achieved through engineered systems, not only rules
- Clinical engineers play a key role in design, selection, and maintenance of safety-critical technologies
- Focus: department-specific technical safety systems

What Is Hospital Safety from an Engineering View?



What Is Hospital Safety from an Engineering View?

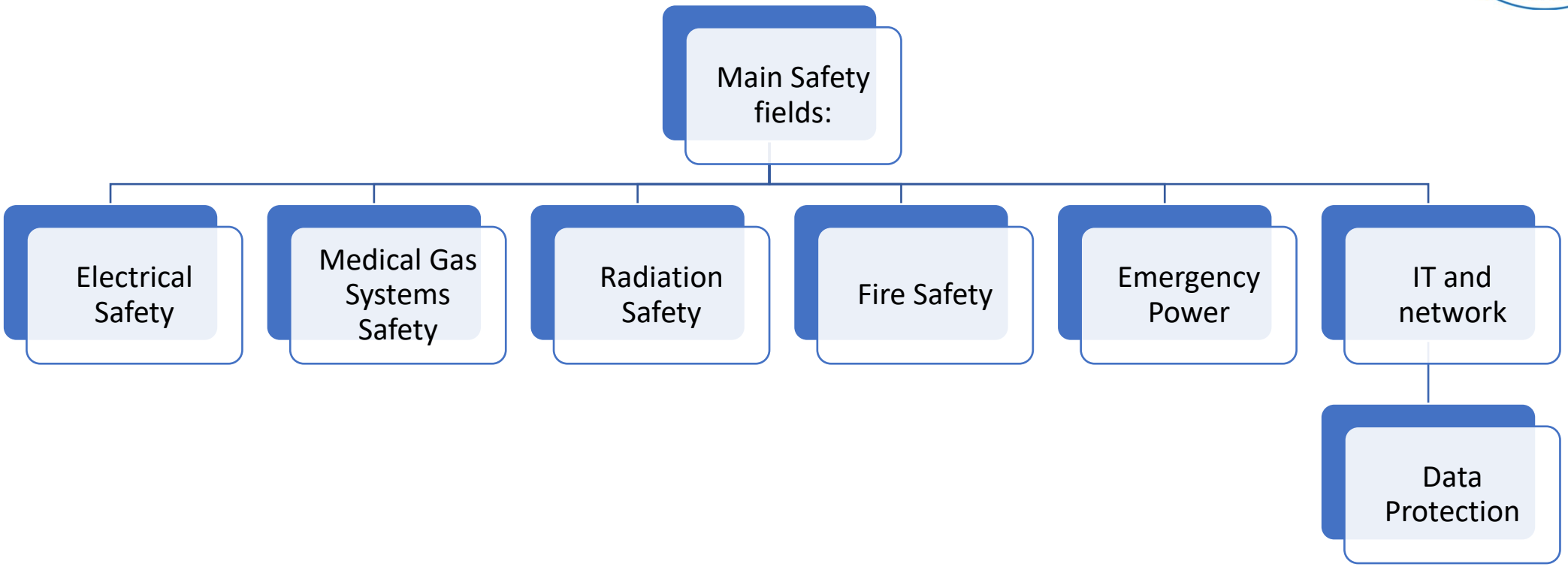
- Achieved by:
  - Technical infrastructure
  - Certified medical devices
  - Monitoring and alarm systems
  - Preventive maintenance programs

What Is Hospital Safety from an Engineering View?

- Safety failures are often caused by:
  - System design problems
  - Equipment malfunction
  - Human–machine interaction errors

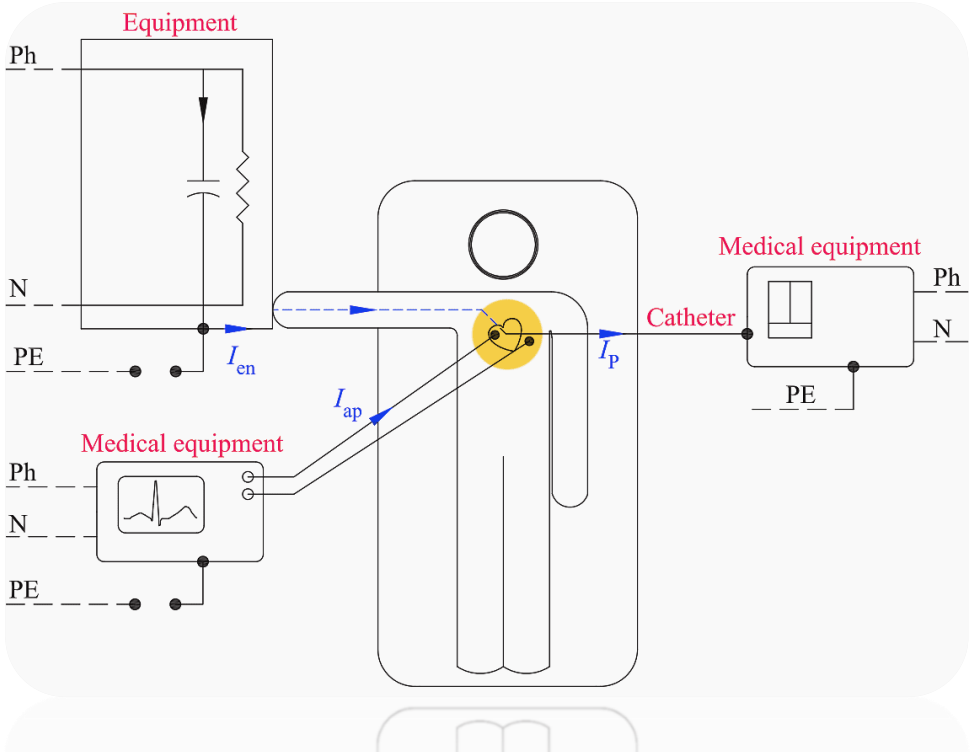
Role of Clinical Engineering in Hospital Safety

- Evaluation and selection of safe medical devices
- Acceptance testing after installation
- Preventive maintenance and calibration
- Incident investigation and reporting
- Coordination with:
  - IT department
  - Facility management
  - Radiation safety officers
- Ensures safety across the **entire device life cycle**


Main Safety fields:
Electrical Safety
Medical Gas Systems Safety
Radiation Safety
Fire Safety
Emergency Power
IT and network
Data Protection

1) Electrical Safety Systems in Hospitals

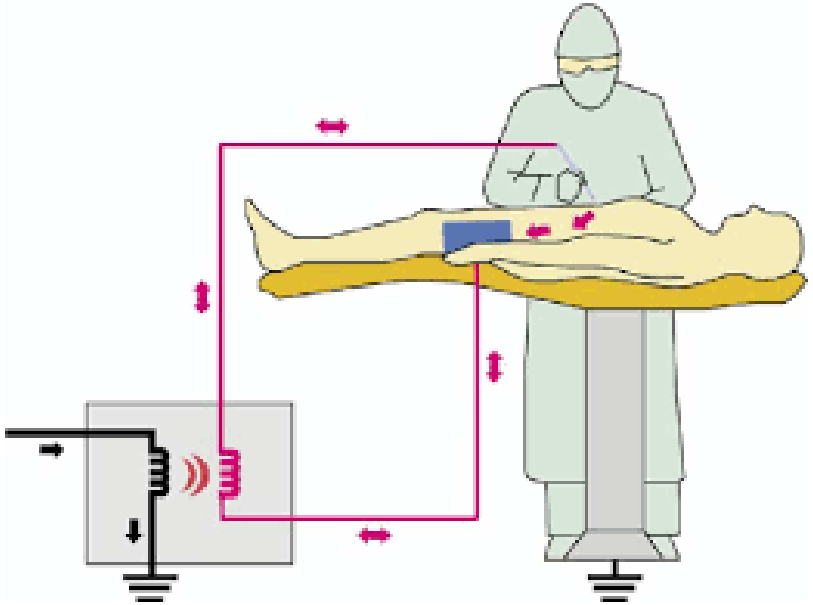
- Major risks:
  - Electric shock to patients
  - Fire caused by faulty equipment
- Engineering safety controls:
  - Grounding (earthing) systems
  - Isolation transformers in critical areas
  - Leakage current protection devices
  - Equipotential bonding in operating rooms



**** This diagram shows how tiny leakage currents from multiple medical devices can pass through a patient—especially via catheters near the heart—and why isolation and equipotential grounding are used to prevent dangerous micro-shock.*

1) Electrical Safety Systems in Hospitals

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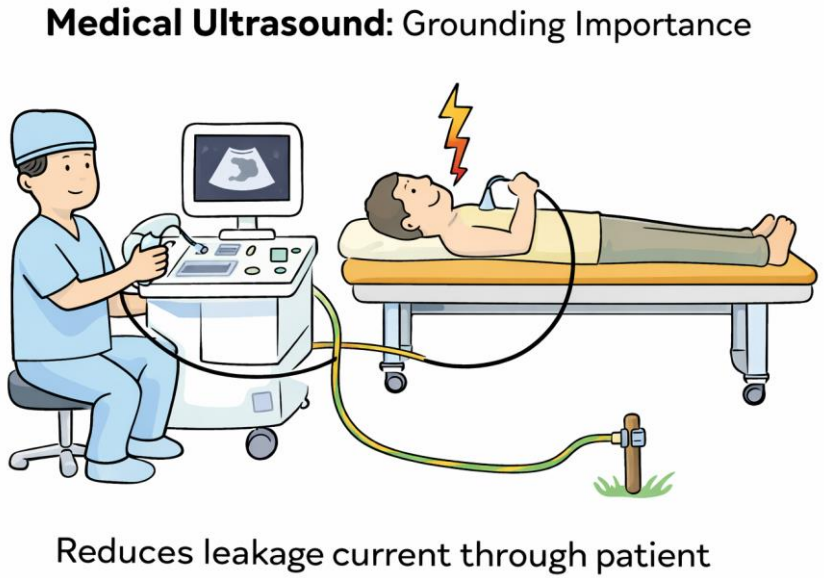


**** This diagram shows an electrosurgical unit where high-frequency current passes from the active electrode through the patient to the return path and ground, highlighting the need for proper grounding and isolation to prevent unintended burns or shock.*

1) Electrical Safety Systems in Hospitals

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**** This diagram shows how poor grounding can introduce electrical noise into an ultrasound system, reducing probe sensitivity and causing image distortion, while proper grounding stabilizes signals and improves image quality.*

Electrical Safety: Clinical Engineering Tasks

- Special locations:
  - ICU
  - Operating rooms
  - Catheterization labs
- Electrical safety testing of devices:
  - Leakage current measurement
  - Ground resistance checks
  - Verification of proper installation
  - Monitoring power quality problems
  - Removing unsafe equipment from service
  - Supporting electrical safety audits

2) Medical Gas Systems Safety

- Medical gases are life-critical utilities, ex.:
  - Oxygen
  - Medical air
  - Nitrous oxide (N2O)
- Engineering safety components:
  - Central supply with backup sources
  - Alarm systems for pressure and flow
  - Zone shut-off valves
  - Standardized connectors and color coding
- Main risks:
  - Gas supply interruption
  - Wrong gas delivery
  - Fire and explosion hazards



Gas	U.S. Color Code	ISO Color Code
Carbon Dioxide	Grey	Grey
He-O_2	Brown & Green	Brown & White
Instrument Air	Red (USA Only)	
Medical Air	Yellow	Black & White
Nitrogen	Black	Black
Nitrous Oxide	Blue	Blue
O_2-He	Green & Brown	White & Brown
Oxygen	Green	White
Vacuum (Suction)	White	Yellow
WAGD (Evac)	Purple	Purple

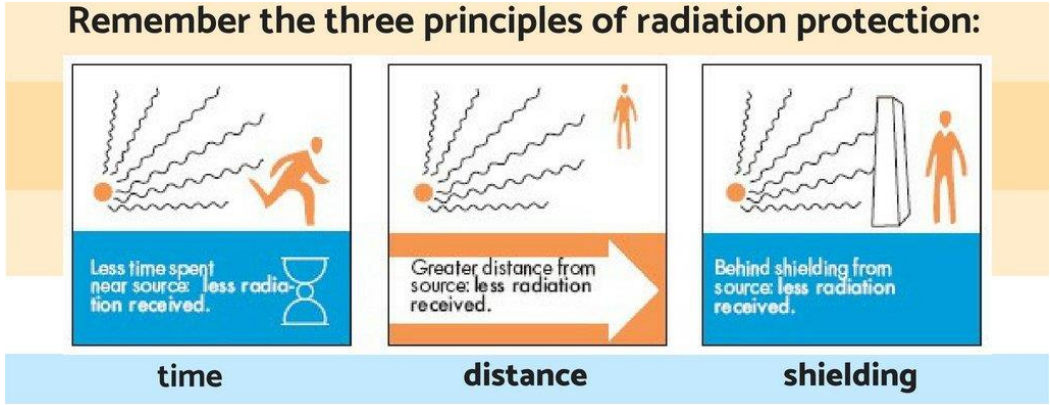
Material Property	Letter Color on Field Color	Example
Single Gases		
Oxygen USP†	White on Green	OXYGEN 50-55 PSI
Carbon Dioxide†	White on Gray	CARBON DIOXIDE
Nitrous Oxide†	White on Blue	NITROUS OXIDE
Cyclopropane‡	Black on Orange	CYCLOPROPANE
Helium USP†	White on Brown	HELIUM 50-55 PSI
Nitrogen NF†	White on Black	NITROGEN 160-200 PSI
Medical Air USP†	Black on Yellow	MEDICAL AIR
Instrument Air†	White on Red	INSTRUMENT AIR
Waste Anesthetic Gas Disposal†	White on Purple	WASTE ANESTHETIC
Laboratory Air†	Black on White/Yellow Checkerboard	LABORATORY AIR
Laboratory Vaccum†	Black on Black/White Checkerboard[2]	LABORATORY VACUUM
Medical-Surgical Vaccum†	Black on White	MEDICAL VACUUM

Material Property	Letter Color on Field Color	Example
Gas Mixtures of Oxygen USP and Nitrogen NF		
19.5% to 23.5% Oxygen‡	Black on Yellow	20% OXYGEN 80% NITROGEN
All other Oxygen Concentrates†‡	Green on Black	40% OXYGEN 60% NITROGEN
Other Gas Mixtures		
Oxygen USP and Nitrous Oxide USP[1†‡]: 52.5% Oxygen USP, 47.5% Nitrous Oxide USP	Defined by User	52.5% OXYGEN 47.5% NITROUS OXIDE
Oxygen USP and Carbon Dioxide USP[1†‡]: ≥ 93% Oxygen USP, remainder Carbon Dioxide USP	Gray on Green	93% OXYGEN 7% CARBON DIOXIDE
Oxygen USP and Helium USP[1†‡]: ≥ 20% Oxygen USP, remainder Helium USP	Green on Brown	26% OXYGEN 74% HELIUM

† Complies to NFPA 99 color marking standards.
‡ Complies to CGA C-9 color marking standards.
1 Use the background colors of the Single Gases. For other gas mixture pipe labels, use the major gas for label color and use the minor gas color for text color.
2 Use a white box with a black outline behind the text.

3) Radiation Safety in Medical Imaging / Therapy Departments

- Areas with radiation exposure:
  - X-ray
  - CT
  - Fluoroscopy
  - Nuclear medicine



- Safety must rely on **engineering controls**, not only procedures
- Main goal:
  - Limit dose to patients, staff, and public

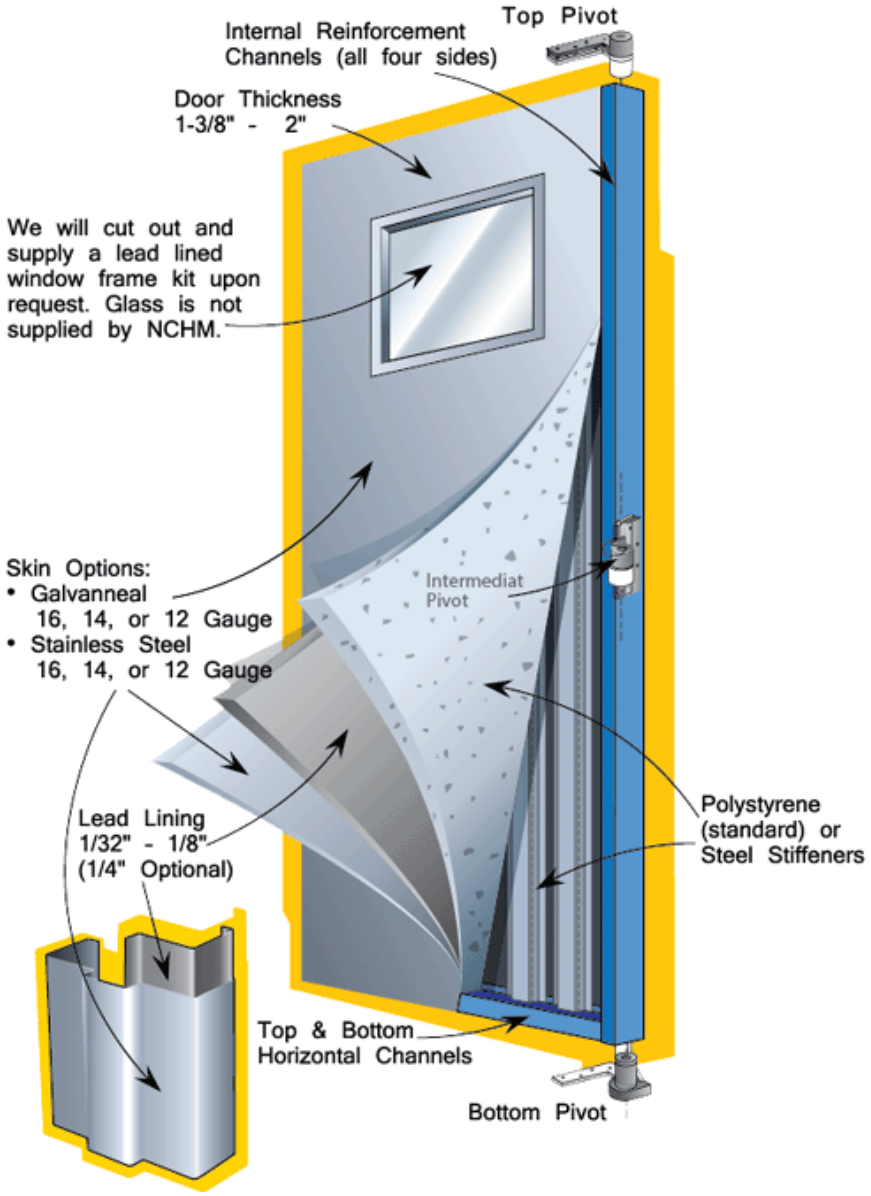
Radiation Safety Engineering Controls

- Structural shielding:
  - Lead-lined walls and doors
- Interlock systems:
  - Radiation stops if door opens
- Warning systems:
  - Light indicators
  - Audible alarms
- Area radiation monitors
- Controlled access zones



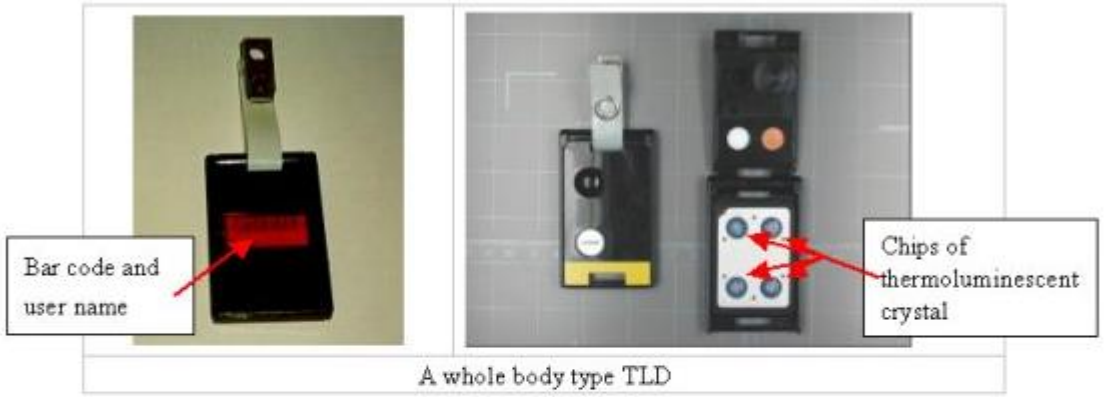
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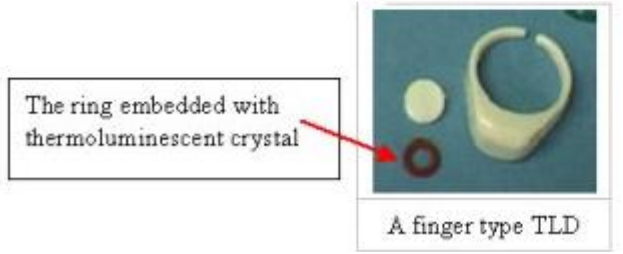


Clinical Engineering Role in Radiation Safety

- Acceptance testing after installation
- Routine quality control (QC) testing
- Verification of safety interlocks
- Dose monitoring system checks
- Cooperation with radiation safety officers and physicists



A whole body type TLD



A finger type TLD



4) Fire Safety Engineering in Hospitals

- Hospitals have high fire risk due to:
  - Oxygen-rich environments
  - Electrical equipment
  - Limited patient mobility
- Engineering-based safety systems:
  - Smoke detectors
  - Fire alarm networks
  - Automatic sprinkler systems
  - Fire-resistant compartment design



5) Evacuation and Emergency Power Systems

- Emergency power supplies for:
  - Ventilators
  - Monitors
  - Infusion pumps
- Backup generators and UPS systems
- Emergency lighting and exit signage
- Fire compartmentation to slow fire spread
- Evacuation plans supported by:
  - Physical building design
  - Automated alerts

Equipment Alarms and Fail-Safe Mechanisms

- Safety-critical devices must detect:
  - Sensor failure
  - Power failure
  - Therapy delivery errors
- Engineering protections:
  - Redundant sensors
  - Automatic shutdown modes
  - Battery backup systems
- Purpose:
  - Prevent harm before clinical staff can intervene

6) IT and Network Safety in Modern Hospitals

Many medical devices are connected to hospital IT systems

- Patient monitors
- Infusion pumps
- Imaging systems (PACS)
- Electronic Health Records (EHR)

Purpose of connectivity:

- Real-time monitoring
- Faster clinical decisions
- Centralized data storage

But this also increases system dependency and risk

Cybersecurity Risks and Their Impact on Care

Common IT risks:

- Unauthorized access to devices or systems
- Malware and ransomware attacks
- Data manipulation or loss

Possible consequences:

- Therapy interruption
- Incorrect clinical information
- Department or system shutdown
- Threats to patient safety

Engineering Controls & Clinical Engineering Responsibilities

Technical safety measures:

- Network segmentation for medical devices
- Secure configuration and authentication
- Regular updates and patch management
- Data backup and disaster recovery systems

Role of Clinical Engineers:

- Risk assessment of connected devices
- Approval and testing of new equipment
- Coordination with IT teams
- Incident reporting and response planning


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Why Standards and Regulations Are Critical for Safety

- Safety cannot rely only on hospital practices
- Devices and systems must be safe before entering the hospital
- Standards define:
  - How devices are designed
  - How risks are controlled
  - How performance is tested
- Regulations ensure:
  - Only compliant devices reach the market

Role of Technical Standards in Medical Safety

- International standards provide:
  - Design requirements
  - Testing procedures
  - Safety limits
- Major organizations:
  - IEC (International Electrotechnical Commission) — electrical and electronic safety
  - ISO (International Organization for Standardization) — risk and quality management
  - NFPA (National Fire Protection Association) — fire protection systems
  - AAMI (Association for the Advancement of Medical Instrumentation) — medical device performance guidelines

How Standards Directly Improve Patient Safety

- Examples:
  - IEC 60601 → limits leakage current in medical devices
  - NFPA 99 → medical gas system safety requirements
  - ISO 14971 → structured risk management process
- Results:
  - Reduced electrical shock risk
  - Safer gas delivery systems
  - Identification of device failure modes

Certification and Regulatory Approval

- Devices must pass:
  - Safety testing
  - Performance evaluation
  - Risk assessment
- Examples of regulatory systems:
  - CE marking (European Union) (for medical devices)
  - FDA clearance or approval (USA)
  - MDR compliance (EU regulation)

Regulatory Approval as a Safety Barrier

- Regulatory pathways require:
  - Clinical evaluation
  - Manufacturing controls
  - Post-market surveillance
- Prevents:
  - Unsafe or ineffective devices from being sold
- Hospitals must not use:
  - Non-certified medical equipment

ISO 13485 and Quality Management Systems

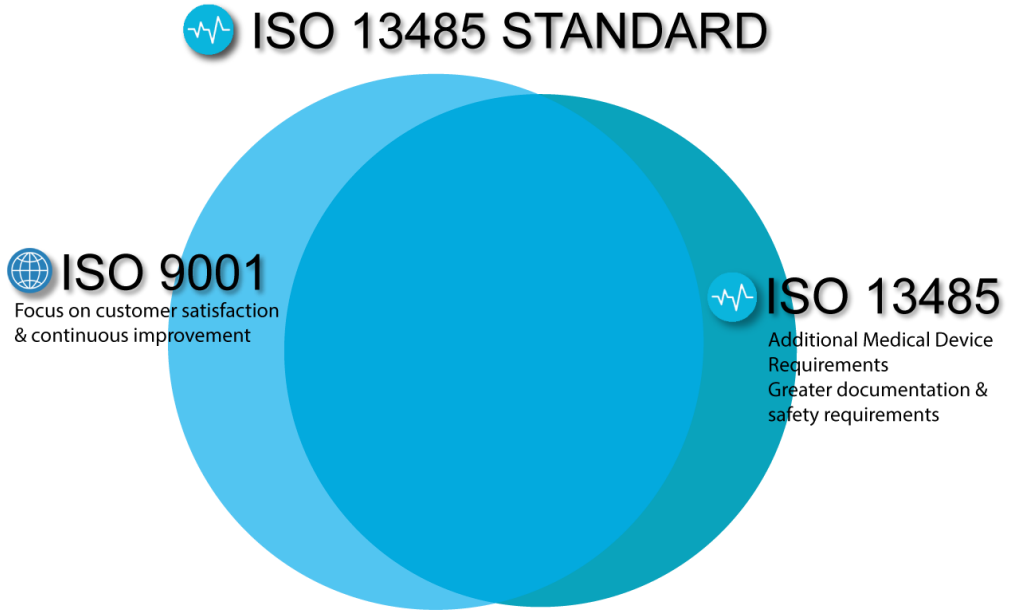
- ISO 13485 focuses on:
  - Medical device manufacturing quality
  - Process control
  - Traceability and documentation
- Quality Management Systems (QMS) ensure:
  - Consistent production
  - Control of changes
  - Corrective and preventive actions (CAPA)



ISO 13485 is a medical-device-specific standard that defines how a Quality Management System (QMS) must be implemented.

QMS , ISO 9001 and ISO 13485

- QMS is used by both medical manufacturers and hospitals.
- ISO 9001 is a general QMS standard (can be used by hospitals).
- ISO 13485 is the QMS standard specifically for medical device manufacturers.



Hospital Responsibility in Regulatory Compliance

- Hospitals must ensure:
  - Only approved devices are purchased
  - Devices are used according to intended purpose
- Risks of non-compliance:
  - Increased patient harm
  - Legal liability
  - Loss of accreditation
- Compliance is part of:
  - Patient safety programs

Clinical Engineering in Device Procurement and Safety

- Clinical engineers evaluate:
  - Certification and conformity documents
  - Technical performance
  - Compatibility with hospital systems
- Participate in:
  - Tender and purchasing committees
  - Technical specification writing
- Ensure:
  - Safety requirements are included in contracts

Lifecycle Management as a Safety Strategy

- Device safety must be maintained through:
  - Acceptance testing
  - Preventive maintenance
  - Calibration
  - Software updates
- End-of-life decisions:
  - Based on safety and performance data
- Poor lifecycle management leads to:
  - Increased failure rates
  - Hidden patient risks

Clinical Engineers as Patient Safety Advocates

- Bridge between:
  - Clinical staff
  - IT departments
  - Facility engineering
  - Manufacturers
- Translate technical risks into:
  - Clinical safety actions
- Support:
  - Incident investigations
  - Root cause analysis
  - System improvement plans

Think with me !? 🤔

1. In the context of hospital safety, what does Root Cause Analysis (RCA) mean?
2. Should clinical engineers be involved in investigations when serious incidents occur, such as patient injury or death in the ICU or after surgery? Why?


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Thank you